

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

NAME:PREETHI.R

REG NO:192511352

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I PREETHI.R(192511352)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Preethi.R

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Article Review: Machine Learning for Heart Disease Diagnosis

Article Details

Authors	Waleed Alsabhan & Abdullah Alfadhly
Title	Effectiveness of machine learning models in diagnosis of heart disease: a comparative study
Journal	Scientific Reports (Springer Nature), Vol. 15, Article 24568
Publisher / Database	Springer (Scientific Reports is a Springer Nature journal)
Year	2025 (Published 8 July 2025)
DOI	https://doi.org/10.1038/s41598-025-09423-y
Open Access PDF	https://www.nature.com/articles/s41598-025-09423-y.pdf
Domain / Application Area	Healthcare and clinical decision support — cardiology diagnostics
Specific ML Problem	Binary classification of heart disease presence/absence from tabular clinical attributes, benchmarking 12 supervised models including Decision Tree, Logistic Regression, Naive Bayes, and SVM

Task 1: Article Summary

(a) Citation, Domain and ML Problem

Full citation: Alsabhan, W., & Alfadhly, A. (2025). Effectiveness of machine learning models in diagnosis of heart disease: a comparative study. *Scientific Reports*, 15, 24568. <https://doi.org/10.1038/s41598-025-09423-y>

This paper sits in the healthcare / clinical decision-support space, specifically cardiology. The problem the authors are solving is a pretty classic binary classification task: given a patient's clinical data (age, blood pressure, cholesterol, chest pain type, ECG readings, and so on), predict whether or not they have heart disease.

What makes it relevant to this course is the range of algorithms they compare. They run Decision Tree, Logistic Regression, Naïve Bayes, and SVM side by side with several ensemble and neural approaches (Random Forest, Gradient Boosting, LightGBM, CatBoost, AdaBoost, KNN, LDA, and a small ANN). So instead of reading about one algorithm in isolation, you actually get to see how a Decision Tree and the statistical-learning models stack up against each other on the exact same data.

(b) Key Objective and Research Gap (150–200 words)

Honestly, the paper isn't trying to invent a new algorithm — its real goal is to figure out which existing model, combined with which preprocessing choices, gives the most reliable heart-disease prediction. To do that, the authors take twelve different ML/DL models and run each one under four different feature-scaling setups (no scaling, standardisation, min-max, and normalisation), while also using SMOTE to fix the class imbalance in the dataset and MI/PCA analysis to figure out which features actually matter.

The gap they point out in the literature is fairly simple but fair: most earlier heart-disease-prediction papers only test three to six algorithms, and almost none of them look closely at how the choice of feature scaling changes the outcome. So there wasn't really a single study you could point to and say 'here's how all these models compare under the same conditions.' The authors fill that gap by testing everything together — same dataset, same splits, seven different evaluation metrics (accuracy, precision, recall, F1, AUC, Cohen's Kappa, log loss) — so the comparison is actually apples-to-apples instead of pieced together from a dozen unrelated papers.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset

Decision Tree (DT): This one is a non-parametric supervised classifier that builds a tree — each internal node tests an attribute, each branch is an outcome of that test, and each leaf is a predicted class. As far as I can tell, the authors just used a standard CART-style tree here; there's no mention of pruning or depth limits, and honestly that shows up later in the results.

Statistical Learning models: Logistic Regression predicts the probability of a class using the sigmoid function and cuts it off at 0.5 — a linear, probability-based classifier at heart. Naïve Bayes leans on Bayes' theorem and just assumes (naïvely, hence the name) that all the features are independent of each other. SVM tries to find the widest possible margin between classes using a separating hyperplane, and can use kernel tricks when the data isn't linearly separable.

Dataset: They used a public UCI Machine Learning Repository dataset that's actually a merge of four separate heart datasets — Cleveland, Hungary, Switzerland, and VA Long Beach — combined on 14 shared attributes like age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG results, max heart rate, exercise-induced angina, and ST depression. This merged dataset shows up a lot in the literature and usually has somewhere around a few hundred patient records, though the version of the paper I could access didn't restate the exact row count — worth double-checking against the paper's Table 1 if your instructor wants an exact figure.

Data preparation: Categorical fields were label-encoded with scikit-learn, and class imbalance was handled with SMOTE, which generates synthetic minority-class examples instead of just copying existing ones. They also ran Pearson correlation and Mutual Information scoring to see which features actually mattered, used PCA to pull out the important ones, and tested four different scaling setups — no scaling, standardisation, min-max, and normalisation — each with an 80:20 train/test split.

(b) Mapping to CO4 and Strength/Limitation

CO4 mapping: The Decision Tree piece maps straight onto the Decision Trees topic — tree induction, splitting criteria, interpretability, all of that. Logistic Regression, Naïve Bayes, and SVM map onto Statistical Learning Methods. And really, the whole idea of the paper — learning a general classification rule from a bunch of labelled patient examples — is basically Inductive Learning in action.

Methodological strength: What I liked most is just how thorough the comparison is. Testing 12 models across 4 preprocessing setups with 7 different metrics gives a much fuller picture than the usual 3-6-model paper, and it actually shows something useful: distance/gradient-based models (SVM, KNN, LR, ANN) care a lot about feature scaling, while tree-based models (DT, RF, GBM) barely notice it. That's a genuinely transferable takeaway, not just a one-off result.

Methodological limitation: On the flip side, there's no mention of hyperparameter tuning anywhere — no pruning depth for the tree, no kernel/C tuning for SVM, nothing. That makes me a bit skeptical of the Decision Tree's weak showing; its log loss is wildly high (around 6.7-7.1, way worse than every other model), which really looks like a symptom of an unpruned, poorly calibrated tree rather than a fair test of what a Decision Tree can actually do. It's also unclear whether SMOTE was applied before or after the train/test split — if it was done before, synthetic samples built from test-set neighbours could leak into training and quietly inflate every reported number.

Task 3: Results & Critical Evaluation

(a) Key Results

Here's a summary of the numbers reported for the no-scaling condition (this is Table 2 in the paper, which the authors treat as their baseline). A couple of values are flagged below because the paper itself rounds or skips them in places.

Model	Accuracy	Precision	F1-Score	Cohen's Kappa	Log Loss
Logistic Regression (LR)	0.8480	0.8480*	0.8480	0.6898	0.3781
Naïve Bayes (NB)	0.8824	0.8822	0.8822	0.7595	0.5216
Decision Tree (DT)	0.8039	0.8039	~0.80	~0.60 (mod.)	7.0674
Support Vector Machine (SVM)	0.8676	0.8701	0.8662	0.7261	N/A
Random Forest (RF)	~0.87–0.89	~0.87–0.89	~0.87–0.89	>0.73	Low
K-Nearest Neighbours (KNN)	0.8725	0.7366	~0.80	1.1207**	—
Gradient Boosting Machine (GBM)	>0.90 (best overall)	>0.90	>0.90	0.7991 (highest)	0.27–0.31
LightGBM (LGBM)	0.8824	0.8822	0.8822	0.7595	0.2801
CatBoost (CB)	0.8971	0.8973	~0.897	0.7887	0.2734–0.2735 (lowest)
AdaBoost / LDA / SGD / RC	0.8186–0.8431	—	—	moderate–substantial	—

*LR and ANN happened to land on the exact same F1 score here. **One thing worth pointing out — KNN's reported Cohen's Kappa of 1.1207 is actually impossible, since Kappa maxes out at 1.0. That's almost certainly a typo somewhere in the original paper, but it's the number as printed.

Best overall model: Gradient Boosting won out on raw accuracy — 90.20% under min-max scaling, the best of any model. CatBoost was the most well-calibrated in terms of probability estimates (lowest log loss at 0.2734, under standardisation) and also had a solid Cohen's Kappa of 0.7887. LightGBM didn't top any single category but stayed strong and consistent no matter which scaling method was used, which honestly makes it a pretty safe default choice.

(b) Critical Evaluation of Results

Are these numbers realistic? For the most part, yes — accuracies in the 80-90% range line up with what you'd expect from similar UCI-based heart-disease studies, so nothing here screams 'too good to be true.' That said, a few things made me pause while reading:

- It's not clear whether SMOTE was applied before or after splitting the data into train/test sets. If it was done first, synthetic samples could end up leaking information from the test set into training, which would quietly inflate every metric in the paper.
- Everything is tested on one merged UCI dataset. There's no outside dataset used to check whether the models still hold up on a different population or a different hospital system, so it's hard to say how well any of this generalises.
- The gap between the top models isn't tested for statistical significance — GBM's 90.20% vs CatBoost's 89.71% might be a real difference or might just be noise. We're not told either way.

- The impossible KNN Kappa value (over 1.0) suggests something went unchecked during review, which makes me a little more cautious about trusting every other number at face value too.
- Since there's no hyperparameter tuning described, the comparison might quietly favor models that just work well 'out of the box' (like the boosting methods) over ones that really need tuning to perform, like DT and SVM.

If I were extending this, I'd want to see k-fold cross-validation with confidence intervals reported, testing on a completely separate dataset (something like MIMIC-III/IV, or data from a different country), proper hyperparameter search for every model — especially DT and SVM, since they seem shortchanged here — a leakage-safe pipeline where SMOTE and scaling are only ever fit on the training fold, and some kind of subgroup breakdown by age or sex to check the models aren't quietly worse for some groups of patients.

(c) Real-World Applicability

Where it could be deployed: This kind of model fits naturally as a screening or triage aid in primary care, cardiology outpatient clinics, or telehealth settings where a full diagnostic workup like angiography isn't immediately available. In practice, doctors often lean toward the more interpretable models — Decision Trees or Logistic Regression — even if they're not quite as accurate as GBM or CatBoost, simply because you can actually explain the reasoning to a patient or a colleague.

Challenges in scaling to industry:

- Data availability — the dataset here is old and mostly US/European in origin. Rolling this out somewhere with a different population (different genetics, different comorbidity patterns, different recording habits) risks the model just not generalising well.
- Compute isn't really the bottleneck — these are lightweight tabular models, nothing like training a large image or genomic deep-learning model, so hardware cost shouldn't be a blocker.
- Ethics and regulation are the real hurdles: getting cleared as medical software, auditing for bias across age/sex/ethnicity groups, handling patient data under privacy laws like HIPAA or GDPR, and making sure the tool supports a doctor's judgment instead of replacing it.
- And practically speaking, it would need to plug into existing EHR systems and clinical workflows, plus someone would need to keep monitoring it over time in case patient patterns shift and the model's performance starts to drift.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

Insight 1 — a single Decision Tree has real limits: Seeing the DT consistently trail behind the ensemble methods, and post that huge log loss, really drove home something we talked about in class — a single decision tree is easy to read and explain, but on its own it tends to overfit or just fail to generalise well. It's basically the textbook motivation for why Random Forest and Gradient Boosting exist in the first place: take a weak, overfitting learner and combine many of them into something sturdier.

Insight 2 — scaling matters depending on the model, not universally: I hadn't fully appreciated before reading this how differently models react to feature scaling. SVM, Logistic Regression, and KNN clearly cared about it, while the tree-based models (DT, RF, GBM) barely budged. That makes sense once you think about why — trees split on raw feature values one at a time, while distance- and gradient-based methods are sensitive to the scale of every feature at once. It's a nice concrete example of how a model's underlying assumptions decide what preprocessing you actually need.

Insight 3 — the 'best' model depends on what you're measuring: Depending on whether you care about accuracy, F1, Kappa, or log loss, a different model comes out on top — GBM, CatBoost, or LightGBM could each claim the win depending on the metric. That's a good reminder that no single number tells the

whole generalisation story, and in something like medical diagnosis, the cost of missing a real case is usually much higher than the cost of a false alarm, so the 'right' metric to optimise for isn't always accuracy.

(b) Proposed Extension

What I'd try: Two things, really. First, test the best models (GBM and CatBoost) on a completely different dataset — something like MIMIC-III/IV or a cardiology registry from a different region — just to see if the rankings actually hold up outside this one merged UCI dataset. Second, run SHAP analysis on the Decision Tree and the top-performing ensemble model so their predictions are actually explainable to a clinician, and add a fairness check that reports accuracy and recall separately for different age groups and for men vs. women.

Why it's doable: None of this requires anything exotic. MIMIC-III/IV is freely available to researchers through PhysioNet once you're credentialed, the models here are all lightweight enough to run without needing serious compute, SHAP has solid, well-documented Python support for every model type used in this paper, and the fairness breakdown is basically just slicing the confusion matrix the authors are already generating — no new infrastructure needed.

Why it would matter: This would directly patch the two weakest spots in the original paper — it only tests one dataset, and it doesn't say anything about explainability or fairness. Doing this wouldn't just be an academic exercise; it would actually tell you whether this approach is trustworthy enough to hand to a real clinic, which is the whole point of doing this kind of research in the first place.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand